

Project Report

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites

1. INTRODUCTION

1.1 Project Overview

UNESCO World Heritage Sites represent places of outstanding cultural and natural importance across the world. These sites preserve human history, biodiversity, architecture, and traditions for future generations. However, understanding large volumes of heritage data through traditional reports can be difficult.

This project develops an interactive Tableau-based visualization system integrated with a Flask web application to analyze UNESCO World Heritage Sites. Users can explore heritage sites using interactive dashboards, maps, charts, filters, and story points to gain meaningful insights about countries, regions, categories, danger status, and inscription trends.

1.2 Purpose

The main purpose of this project is to simplify the analysis of UNESCO World Heritage Sites using interactive data visualizations. The project helps students, researchers, and heritage enthusiasts understand heritage distribution, identify endangered sites, compare countries and regions, and analyze historical trends through an easy-to-use web portal.

2. IDEATION PHASE

2.1 Problem Statement

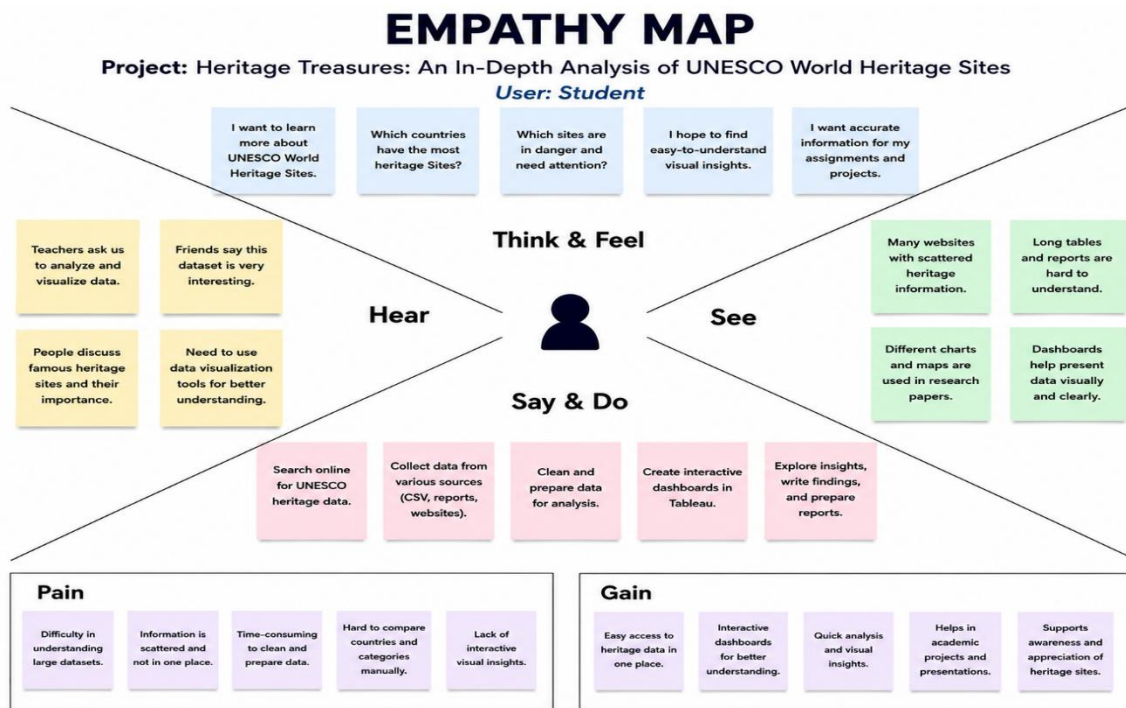
PROBLEM STATEMENT

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites

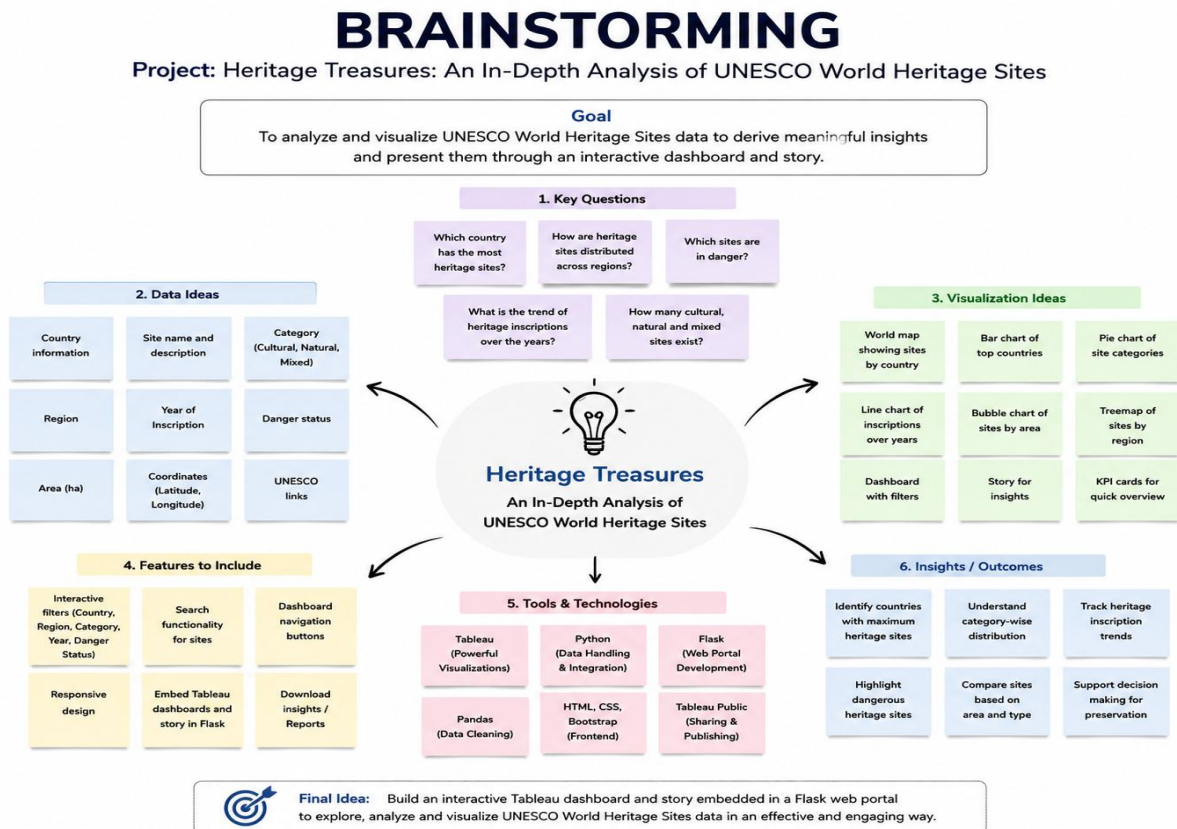
This template outlines the problem and its impact on students who seek meaningful insights about UNESCO World Heritage Sites.

I am  a student who wants to explore UNESCO World Heritage Sites and understand their details, distribution, and significance.	I'm trying to  find reliable and organized information • about heritage sites • across different countries • with easy comparison • through interactive tools	But  it takes a long time • to collect data from multiple sources • to analyze large datasets manually • to identify trends and meaningful patterns	Because  the existing resources • are not interactive • lack visual insights • are scattered and not well-structured • do not support comparative analysis	Which makes me feel  frustrated • with incomplete data • with manual analysis • with time-consuming research • unable to get quick insights
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2.2 Empathy Map Canvas



2.3 Brainstorming



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map



3.2 Solution Requirement

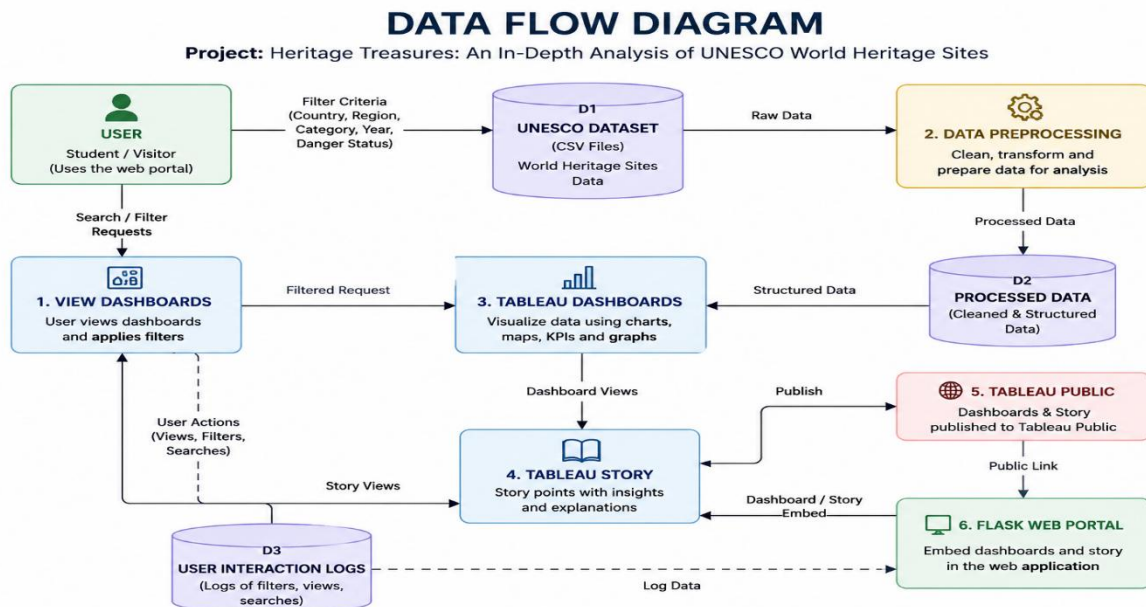
Functional Requirements

- View UNESCO heritage information
- Interactive dashboards
- Tableau Story
- Search and Filters
- Export reports
- Responsive UI

Non-functional Requirements

- Fast response time
- Secure application
- Easy navigation
- Reliable system
- Scalable architecture

3.3 Data Flow Diagram



3.4 Technology Stack

Describe the technologies used.

Technology	Purpose
HTML	Frontend
CSS	Styling
Bootstrap	Responsive Design
JavaScript	Interactivity
Python	Backend
Flask	Web Framework
Tableau	Visualization
GitHub	Version Control
Tableau Public	Dashboard Hosting

4. PROJECT DESIGN

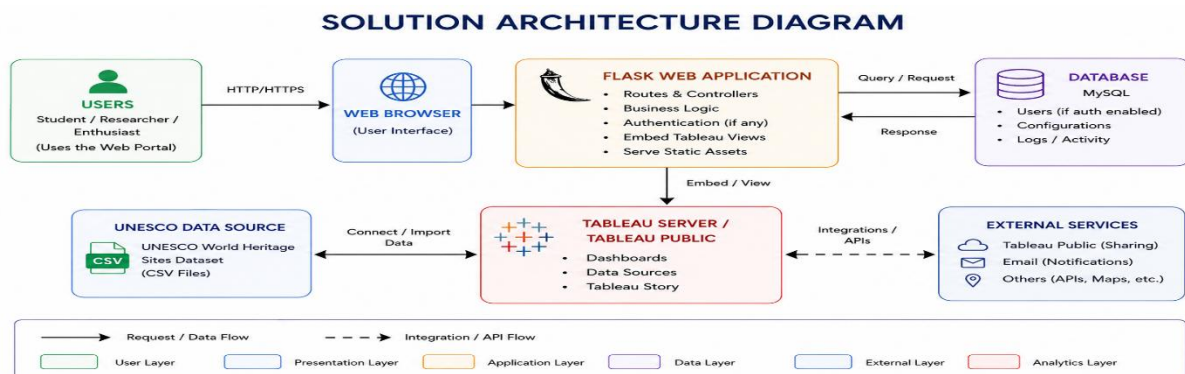
4.1 Problem Solution Fit

PROBLEM – SOLUTION FIT					
Project: Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites					
Define CS, P, Intro CC	1. CUSTOMER SEGMENT(S) Who is your customer? - Students (college & school) - Researchers - Academicians - Heritage enthusiasts - Travelers and general public - Policy makers & organizations Primary User: Student / Researcher	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? - Limited time to analyze large datasets - Lack of data analysis and visualization tools - Difficulty in understanding diverse heritage data - Need for free / low-cost solutions - Dependence on internet connectivity - Access only to static reports or lengthy documents - Limited exposure to data visualization tools	CC	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? - UNESCO official website (limited search & visualization) - Static reports, PDFs, research papers - General data portals (Kaggle, Data.world, etc.) - Spreadsheet analysis (Excel) - Power BI / Tableau Public dashboards - Custom dashboards in other projects Pros: Some visualization available Cons: Not interactive, limited filters, not user-friendly	AS
	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? - Explore and understand UNESCO heritage sites data - Compare countries, categories, regions - Identify sites in danger - Track heritage inscription trends over time - Learn insights through interactive visualizations - Generate reports / insights for academic work	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? - Large and complex heritage data is not easy to analyze - Traditional reports are not interactive - Lack of user-friendly dashboards for quick insights - Limited exposure to data visualization tools - Need for visual storytelling to communicate insights	RC	7. BEHAVIOUR What does your customer do to address the problem and get the job done? - Search for data on websites - Download CSV files and analyze in Excel - Read reports and research papers - Create manual charts - Share findings in presentations / assignments - Spend free time on voluntary work (e.g., awareness)	BE
Focus on J&P, J&P, Intro CC, Intro CC	3. TRIGGERS What triggers customers to act? - Academic assignments / projects - Research requirements - Interest in world heritage & culture - Awareness of data visualization tools - News about heritage sites in danger - Competitions / hackathons	10. YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. An interactive Tableau dashboard and story embedded in a Flask web portal that allows users to explore, analyze and visualize UNESCO World Heritage Sites data with filters (country, region, category, year, danger status), maps, charts, KPIs and Insights. It solves the problem by: - Providing interactive dashboards - Easy filtering and exploration - Visual storytelling for better understanding - User-friendly web interface - Free and accessible solution	SL	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? - Web portal (Flask application) - Embedded Tableau dashboards & story - Search and filter interfaces - Online sharing via links 8.2 OFFLINE What kind of actions do customers take offline? - Academic presentations - Project submissions - Printed reports / screenshots - Discussions and peer sharing	CH
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? Before: Confused, time-consuming, difficult to compare, overwhelmed by raw data After: Informed, confident, insightful, satisfied, excited to explore more				

4.2 Proposed Solution

PROPOSED SOLUTION		
Project: Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites		
S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Students and researchers find it difficult to explore and analyze UNESCO World Heritage Sites data due to lack of interactive and user-friendly tools. Existing solutions are static, non-interactive and do not provide clear insights.
2.	Idea / Solution description	Develop an interactive web portal with embedded Tableau dashboards and story to visualize and analyze UNESCO World Heritage Sites data. Users can filter data by country, region, category, year, and danger status, view maps, charts, KPIs, and export insights.
3.	Novelty / Uniqueness	- Combines powerful Tableau visualizations with a user-friendly Flask web interface. - Provides interactive, filter-based exploration of global heritage data. - Includes a story for data-driven storytelling and easy understanding. - Not just static dashboards – full analytics experience in one portal.
4.	Social Impact / Customer Satisfaction	- Helps students, researchers and academicians in study, projects and presentations. - Promotes awareness about UNESCO heritage sites and conservation. - Saves time in data analysis and report generation. - Users get clear and quick insights leading to better decisions and understanding.
5.	Business Model (Revenue Model)	- Freemium Model – Basic dashboards free for all users. - Premium/Advanced analytics and custom reports for educational institutions, research organizations and tourism boards. - Future scope: API access, data subscription or custom dashboards as paid service.
6.	Scalability of the Solution	- Can be extended with more datasets (e.g., natural heritage, cultural heritage, country-wise data). - Supports real-time updates and additional analytics tools. - Role-based access and user authentication can be added. - Easily deployable on cloud platforms for large user base.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Include:

- Sprint Planning
- Product Backlog
- Story Points
- Sprint Schedule
- Velocity
- Burndown Chart

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Include:

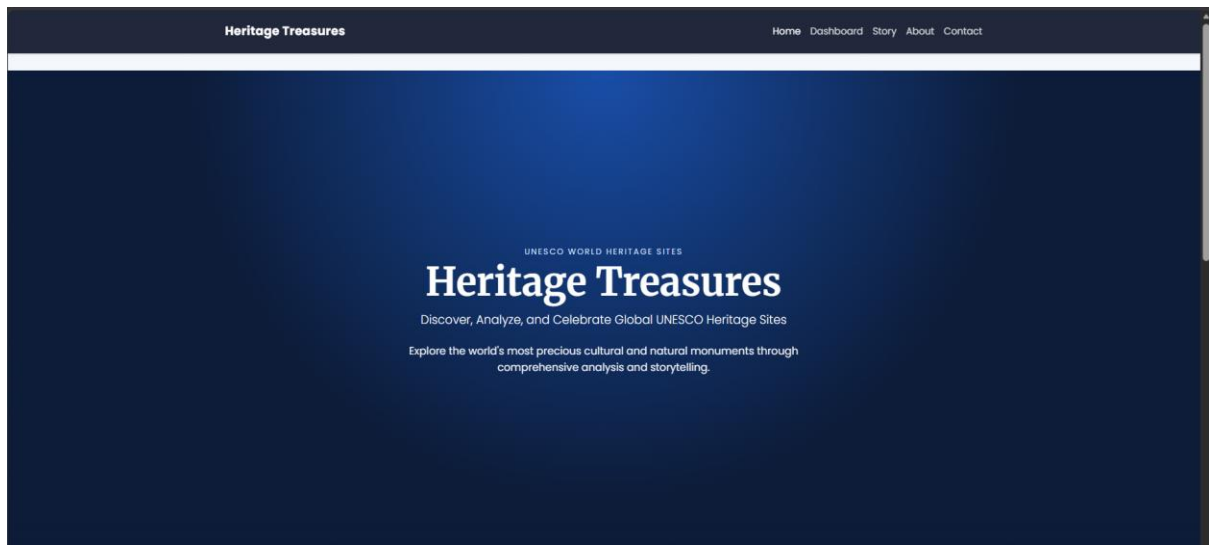
- Dataset Loading
- Data Cleaning
- Filters
- Dashboard Testing
- Story Testing
- Performance Results

7. RESULTS

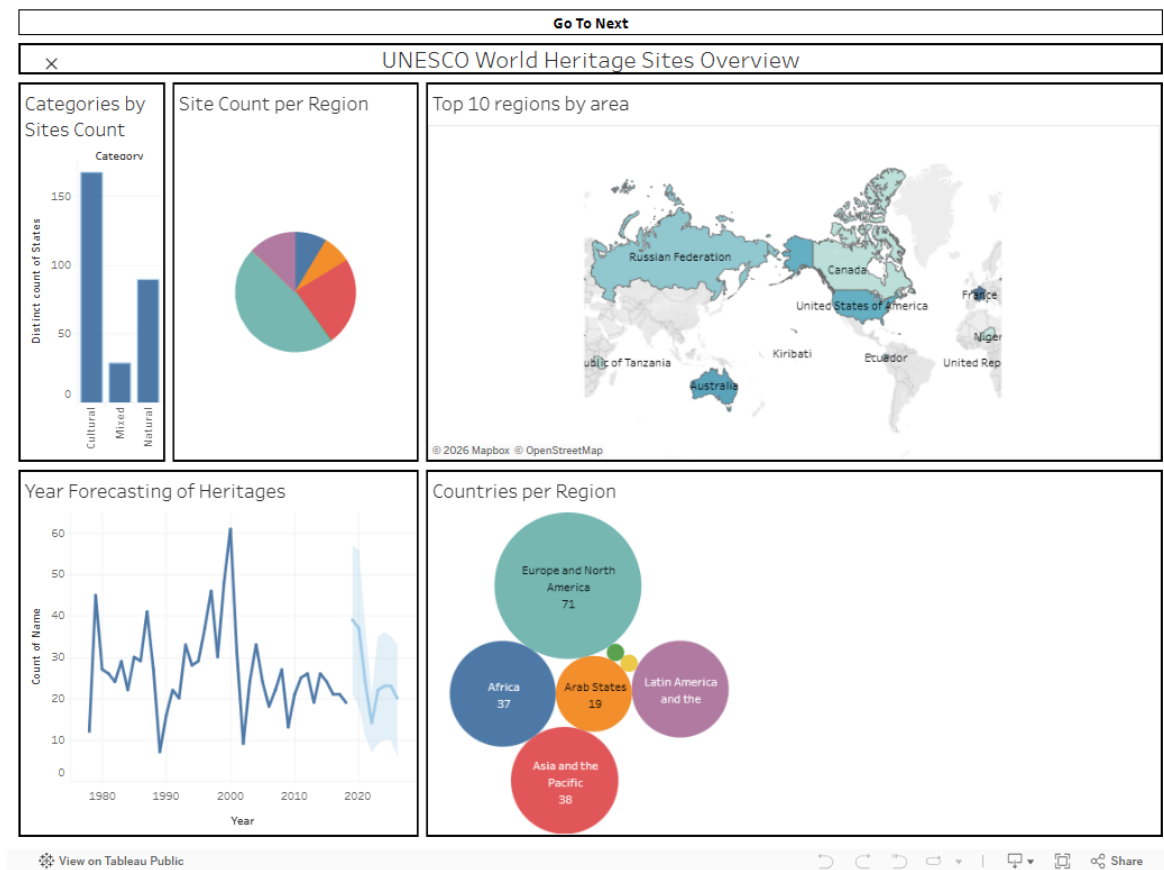
7.1 Output Screenshots

Add screenshots of:

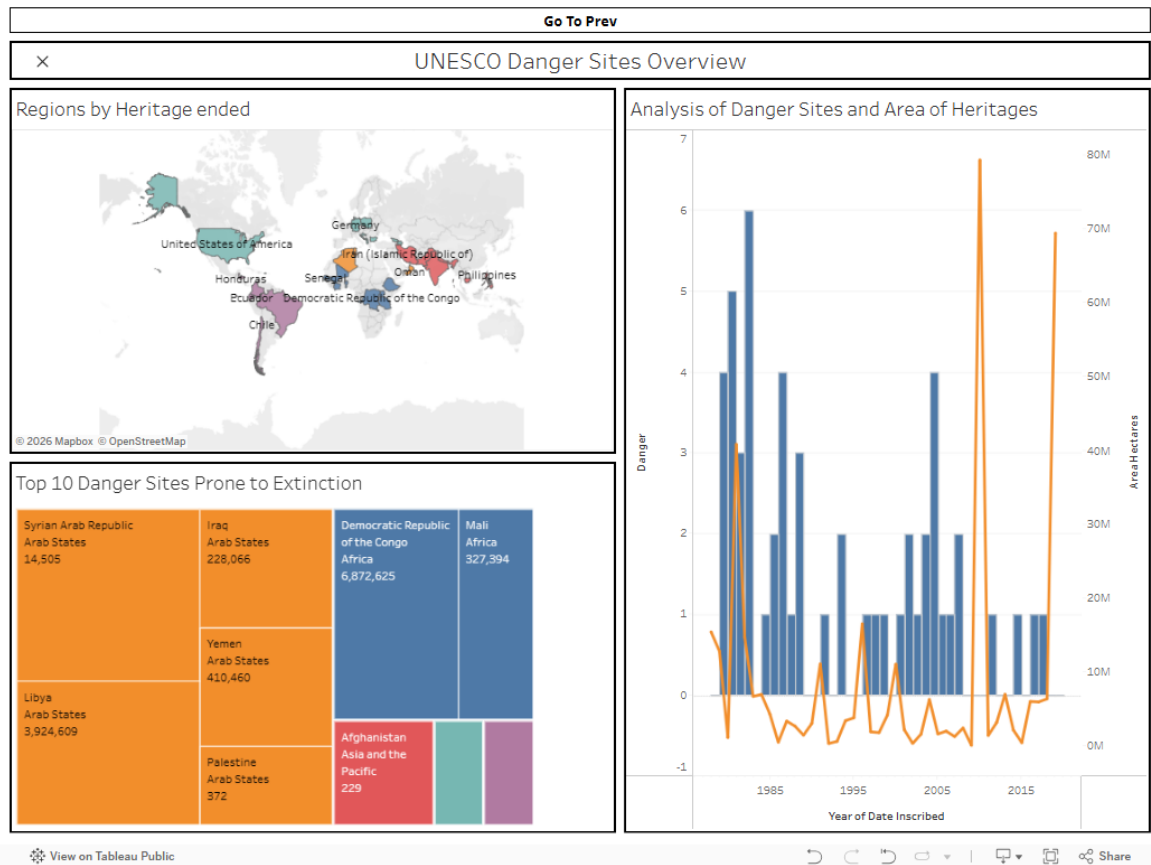
- Home Page



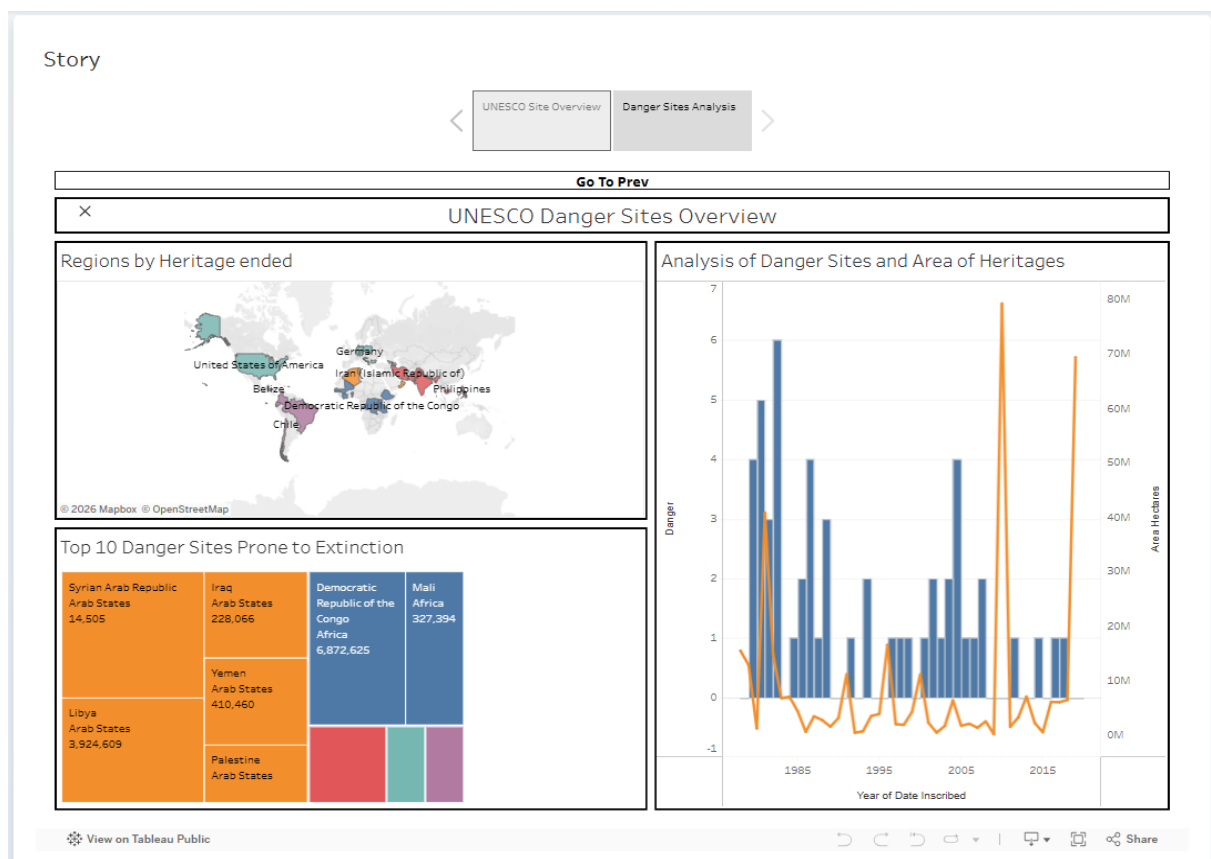
- Dashboard 1



- Dashboard 2



- Tableau Story



8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive visualization of UNESCO data.
- Easy comparison of countries and heritage categories.
- User-friendly interface with filters and dashboards.
- Supports educational and research purposes.
- Embedded Tableau dashboards improve accessibility.
- Helps identify endangered heritage sites.
- Easy deployment through Flask and GitHub.

Disadvantages

- Depends on dataset quality and completeness.
- Requires internet connection for Tableau Public dashboards.
- Limited real-time data updates.
- Large datasets may slightly increase loading time.
- Analysis is limited to available UNESCO data.

9. CONCLUSION

This project successfully demonstrates how Tableau and Flask can be integrated to create an interactive web portal for UNESCO World Heritage Sites analysis. The dashboards and stories provide meaningful insights through maps, charts, KPIs, and filters, making heritage data easier to understand. The project supports education, research, and awareness by transforming complex datasets into interactive visualizations.

10. FUTURE SCOPE

- Real-time synchronization with UNESCO databases.
- AI-based recommendation system for heritage tourism.
- Mobile application support.

- Advanced predictive analytics using Machine Learning.
 - Multi-language support.
 - User authentication and personalized dashboards.
 - Integration with GIS and live mapping services.
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11. APPENDIX

GitHub Repository

[Ksurekha21/Heritage-Treasures-An-In-Depth-Analysis-of-UNESCO-World-Heritage-Sites](https://github.com/Ksurekha21/Heritage-Treasures-An-In-Depth-Analysis-of-UNESCO-World-Heritage-Sites)

Tableau Public Link

Dashboard:

https://public.tableau.com/app/profile/surekha.k7669/viz/HeritageTreasures_17829743037900/UNESCOWorldHeritageSitesOverview?publish=yes

Story:

https://public.tableau.com/app/profile/surekha.k7669/viz/HeritageTreasures_17829743037900/Story?publish=yes

Demo Video Link

https://drive.google.com/file/d/17auJVZ3cfPRhs_Jy8pXKexN6wwTJ6UpR/view?usp=sharing